



3.4.4 Wrap-Up: Reflection

1. Rate your confidence in solving problems using these angle pairs.

I can solve number problems involving....	I am an expert and could help others.	I can hold my own.	I am getting there.	I would like more help or practice.
Congruent Supplements				
Congruent Complements				
Vertical Angles				
Consecutive Interior Angles				
Consecutive Exterior Angles				
Alternate Interior Angles				
Alternate Exterior Angles				
Corresponding Angles				



Unit 3, Lesson 5: Converse of the Alternate Interior Angles Theorem



Benchmarks

MA.912.GR.1.1 Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.

MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all” and “not” statements. Find the converse, inverse, and contrapositive of a statement.



Learning Targets

- ☐ I can write and use the converse of a postulate or theorem.
- ☐ I can use postulates and theorems about parallel lines to determine if lines are parallel.



3.5.1 Warm-Up: Notice and Wonder

The game *Donkey Kong* was released for sale in the U.S. in 1981 by Nintendo, a Japanese playing card and toy company turned video game developer. Mario, circled in yellow, is shown in the image running and climbing up the platforms and ladders.

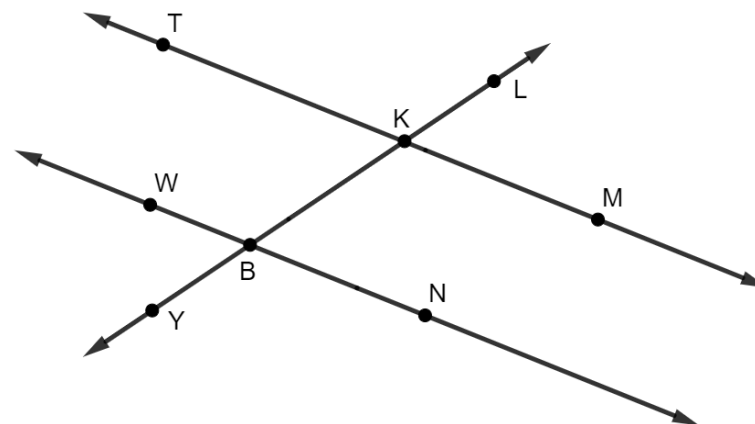


1. What do you notice about the different platforms Mario is running on?
2. What do you wonder?
3. How can you determine if any of the platforms or ladders are parallel?



3.5.2 Exploration: Congruent Angles Formed by Two Lines and a Transversal

1. $\overleftrightarrow{TM}$, $\overleftrightarrow{WN}$, and $\overleftrightarrow{YL}$ are shown. $\overleftrightarrow{YL}$ intersects $\overleftrightarrow{TM}$ and $\overleftrightarrow{WN}$.



Work with your partner to conjecture if the given information in each row of the table is sufficient to show that the lines are parallel.

Angles	Type of Angles	Sufficient?
$\angle TKL \cong \angle WBK$		☐ Yes ☐ No
$\angle MKB \cong \angle WBK$		☐ Yes ☐ No
$\angle MKB \cong \angle TKL$		☐ Yes ☐ No
$\angle TKL \cong \angle YBN$		☐ Yes ☐ No
$\angle WBK$ and $\angle NBK$ are supplementary.		☐ Yes ☐ No
$\angle TKL$ and $\angle WBY$ are supplementary.		☐ Yes ☐ No



3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems

1. Work with your partner or group to write the converse of the postulate or theorem your group is assigned. Then, draw a picture to represent the relationship described.

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.	
Picture	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then corresponding angles are congruent.	
Picture	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.	
Picture	

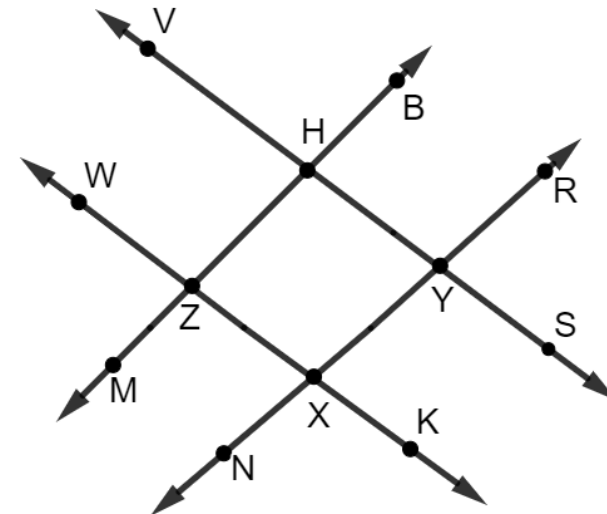
Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.	
Picture	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.	
Picture	



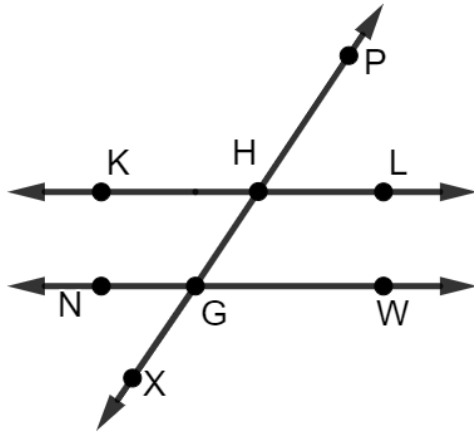
3.5.4 Guided Practice: Determining if Lines Are Parallel

1. $\overleftrightarrow{VS}$, $\overleftrightarrow{WK}$, $\overleftrightarrow{MB}$, and $\overleftrightarrow{NR}$ are shown where $m\angle VHB = 98^\circ$, $m\angle RYS = 78^\circ$, $m\angle NXK = 102^\circ$, and $m\angle WZM = 82^\circ$.



Which pair(s) of lines are parallel? Justify your answer.

2. $\overleftrightarrow{KL}$, $\overleftrightarrow{NW}$, and $\overleftrightarrow{PX}$ are shown.



Determine if the given condition could be used to justify that $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$?	Justification
$\angle PHL \cong \angle WGX$	☐ Yes ☐ No	
$m\angle KHP + m\angle NGX = 180^\circ$	☐ Yes ☐ No	
$\angle KHG \cong \angle WGH$	☐ Yes ☐ No	
$\angle NGX$ and $\angle WGX$ are supplementary.	☐ Yes ☐ No	
$\angle LHG \cong \angle KHP$	☐ Yes ☐ No	



3.5.5 Wrap-Up: Cool Down

- Write a short note to a friend who missed today's class summarizing what you learned.